Assignment 2

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Question 1

(a) The elongation of this constellation is **90 degrees**.
Reason: Since it is midnight, the sun is at the Nadir, and the constellation is at the intersection of the ecliptic and the horizon. Therefore, the elongation is 90 degrees.

(b) Around **sunrise**.
Reason: Fix the position of the constellation and consider the annual motion of the sun on the ecliptic. Since the window faces the east, the sun travels three quarters of the ecliptic in the opposite direction to the daily motion of the celestial sphere. Therefore, the sun arrives at the same place as the constellation, which means he can see the constellation at sunrise.

Question 2

(a) The angular separation is **0-90 degrees (0 and 90 excluded)**.
Reason: Since he saw a crescent Moon, less than half of the moon was lit by the sun from Earth's view. Therefore, the angular separation is less than 90 degrees.

(b) A time between **dawn** and **noon**.
Reason: Fix the constellation on the intersection of the ecliptic and the horizon, keep the angle between the constellation and the sun between 0 and 90 degrees, then the sun's position is between the point where the sun rises and the intersection of the Celestial Meridian and the ecliptic. Therefore, the time is between dawn and noon.

Question 3

After sunset.
Reason: When Mars is in retrograde motion, the sun and Mars are on the opposite side of Earth. Therefore, Mars will rise after sunset.

Question 4

Yes, they will.
Reason: The Gregorian calendar and the Chinese calendar have a 19-year circle. Therefore, after 19 years (which is within their life span), they will celebrate their birthday together again.